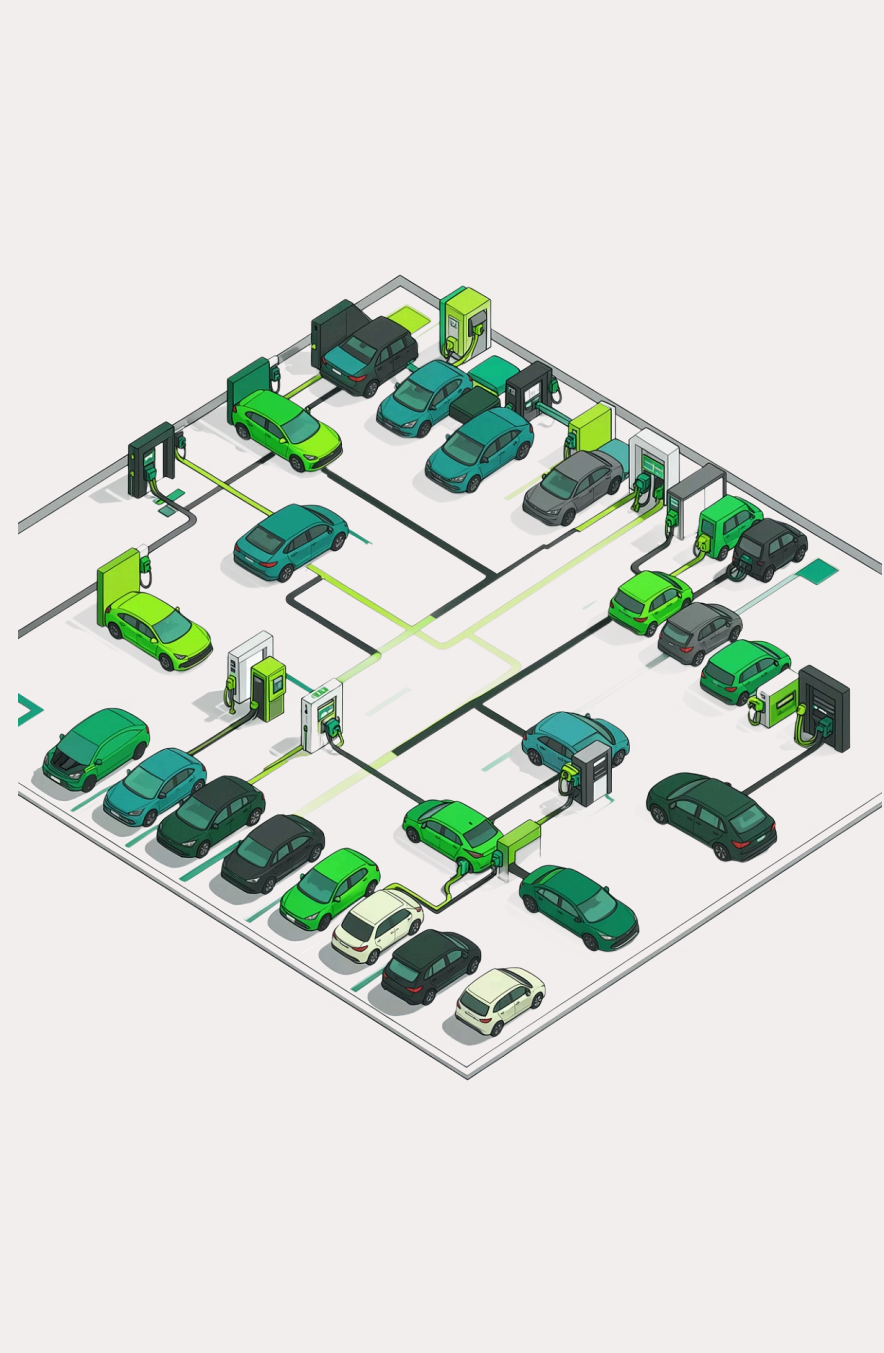




Energy-Constrained Dynamic Travelling Salesman Problem for Electric Vehicle Fleets

TEAM F

TEAM MEMBERS: ANUSHREE K. L, DIPANKAR CHETTRI, VARSHA K B,
and YASHASHWINI PRIYA

Under the guidance of: Dr. S KRISHNA ANAND

Problem Statement

Battery constraints limit EV routing because fleets must operate within finite capacity and account for mandatory mid-route charging, while real-time disruptions such as traffic delays, extended deliveries, and unexpected battery drain can quickly collapse pre-calculated routes. At the same time, a computational bottleneck prevents current software from recalculating energy-feasible routes in real time without causing operational paralysis.

Objectives

01

Dynamic Optimization

E-VRPTW under real-time disruptions

02

Hybrid Algorithmic Approach

Genetic Algorithm sequencing +
Linear Programming energy
calculations

03

Disruption Tolerance

Rolling Horizon Handler patches
routes instantly

04

Modern Dashboard

React/FastAPI platform for operators
and researchers



Existing Systems

Traditional VRP Solvers

Built for combustion engines

- Ignore non-linear EV charging curves
- No battery range anxiety modeling
- Assume unlimited refueling

Static E-VRPTW Solvers

Perfect-day scenarios only

- No disruption tolerance
- Any delay collapses entire schedule
- No real-time rescheduling

Disjointed Tools

Academia meets enterprise

- Academic tools lack operator interfaces
- Commercial dashboards lack algorithmic power
- No integrated solution

Proposed System



Dynamic Real-time Optimization

Continuously re-optimises routes based on live traffic, battery status, and delivery changes, ensuring energy-feasible paths.



Robust Disruption Handling

The Rolling Horizon Handler instantly adapts to unexpected events, maintaining operational continuity and reducing delays.



Intelligent Hybrid Algorithms

Combines Genetic Algorithms for optimal sequencing with Linear Programming for precise energy consumption and charging calculations.



Intuitive Operator Dashboard

A modern React/FastAPI platform offers clear visualisations and controls for efficient fleet management and research insights.

Modules



Data Collection & Acquisition

Ingests benchmark files and real-world API feeds (OMS, FMS, OpenChargeMap, OSRM) into a unified data pipeline.



Genetic Algorithm Engine

Evolves route sequences over thousands of generations using selection, crossover, and mutation operators to find globally optimal delivery schedules.



Stochastic Disruption Engine

Simulates real-world chaos — traffic delays, battery failures, extended service times — and triggers the Rolling Horizon Handler to patch routes without full recomputation.



Core Problem Formulation

Encodes the E-VRPTW into chromosome representations and validates routing sequences against energy and time window constraints.



LP Charging Optimizer

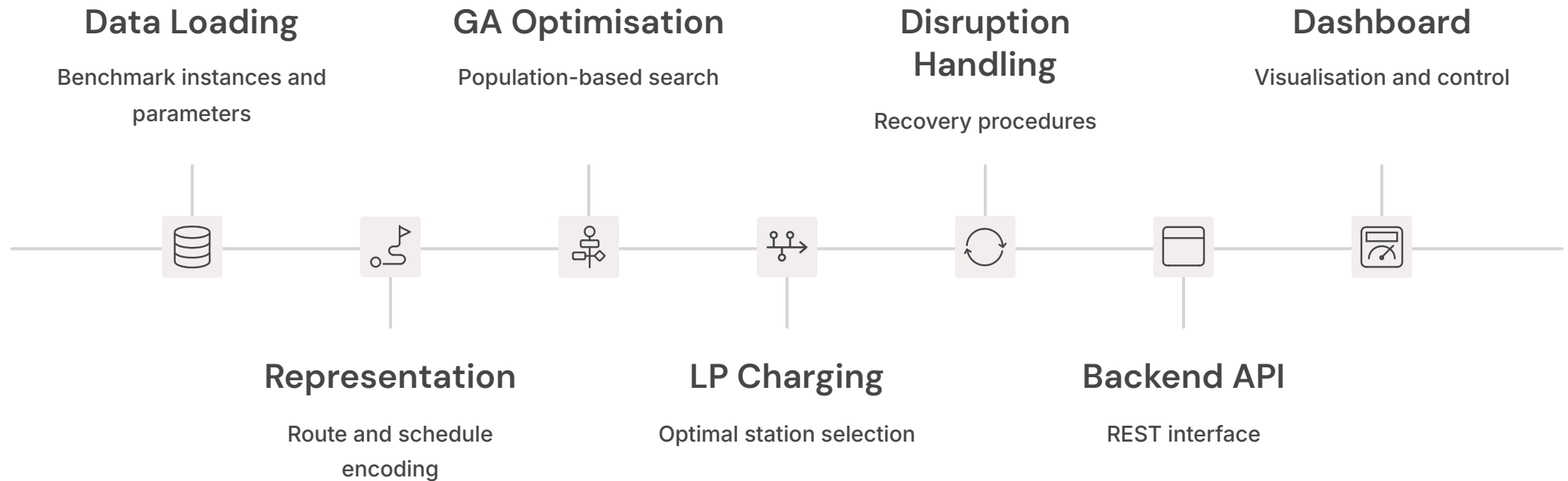
Uses PuLP and HiGHS solvers to mathematically determine the optimal charging stop locations and durations along each route.



Full-Stack Dashboard & Streaming

FastAPI REST backend exposes all algorithmic outputs; React/TypeScript frontend delivers live fleet telemetry via WebSockets and persists state in PostgreSQL.

System Architecture and Workflow



Completed Implementation Phases

01

Data Layer & Benchmark Loading

File parser and instance management

02

Core Problem Formulation

Mathematical models and constraints

03

Genetic Algorithm

Population initialisation and evolution

04

LP Charging Optimisation

Station placement solver

05

Integration & Orchestration

Module coordination

06

Benchmarking & Evaluation

Solution quality metrics

07

Visualisation & Reporting

Dashboard displays

08

REST API Layer

Backend interface

09

Web Frontend

Researcher dashboard

Core Problem Formulation

Decision Space

- Route order and vehicle assignment
- Charging-stop placement
- Schedule feasibility

Objective Function

$$\min Z = \sum_{r \in R} (c_d \cdot d_r + c_q \cdot q_r) + \lambda \cdot P$$

- c_d = cost per unit distance
- d_r = total distance of route r
- c_q = cost per unit charging energy
- q_r = total charging amount on route r
- P = infeasibility penalty
- λ = large penalty weight

Core Problem Formulation (Continued)

Key Constraints

- Battery capacity limits
- Charging feasibility
- Route continuity
- One station per leg maximum
- Operational consistency under disruptions

Battery Feasibility Constraint

$$e_{k+1} = e_k - E_{k,k+1} + Q_{k+1}, \quad 0 \leq e_k \leq Q$$

- e_k = battery level after step k
- $E_{k,k+1}$ = energy used between consecutive nodes
- Q_{k+1} = charging added at that step
- Q = maximum battery capacity

Solution Approach

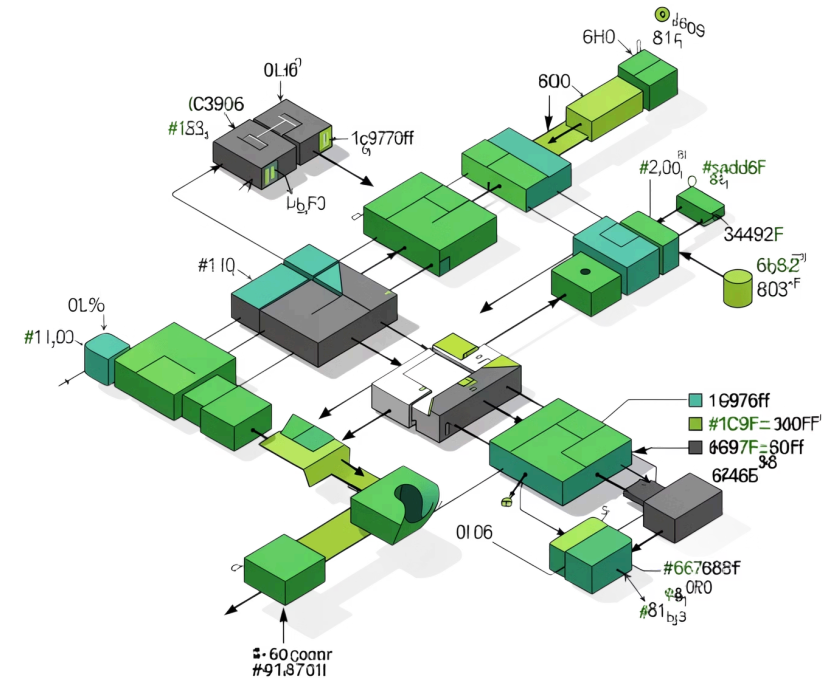
The Genetic Algorithm (GA) explores candidate solutions and the Linear Programming (LP) refines and optimizes charging decisions to ensure feasibility.

Genetic Algorithm Core Logic

Fitness Objective

$$\text{Fitness} = c_{\text{travel}} + c_{\text{charging}} + P_{\text{infeasibility}}$$

Minimise travel cost computed from route distance multiplied by per-kilometre cost, plus charging costs when stations are required, plus large penalties if routes cannot be made feasible.



Genetic Algorithm: Selection, Crossover & Mutation

Selection: Tournament

Competitive selection among random subsets maintains diversity

Crossover: Order Crossover

Preserves customer sequence patterns whilst generating offspring

Mutation: Inversion & Reassignment

Route fragment reversal and vehicle reallocation prevent premature convergence

100

Population Size

Individuals per generation

50

Generations

Evolutionary iterations

0.8

Crossover Rate

Probability of recombination

0.1

Mutation Rate

Genetic variation probability

Linear Programming Charging Optimiser

Objective Function

$$\text{Minimise } \sum_{i,s} (c_s y_{i,s} + d_{i,s})$$

Minimise charging station costs plus detour distance penalties for inserted stops.

Decision Variables

- $y(i,s) \in \{0,1\}$: station s inserted after leg i
- $q(i,s) \geq 0$: charging amount at stop
- $e(k)$: battery level after node k

Constraints

- $q(i,s) \leq Q \cdot y(i,s)$
- $e(k+1) = e(k) - \Delta e + q$
- $e(0) = Q$ (initial charge)
- $\sum_s y(i,s) \leq 1$ per leg

The solver implements energy flow constraints ensuring battery levels remain within feasible bounds. It employs HiGHS as the primary solver, with automatic fallback to CBC if computational resources are constrained.

Researcher Dashboard Overview

01

Instance Upload & Management

Upload benchmark files and view available problem instances

02

Configuration Panel

Set GA parameters, LP solver options, and disruption scenarios

03

Run Control & Monitoring

Launch optimisation jobs and track real-time progress indicators

04

Convergence Analysis

Inspect fitness curves showing best and average population scores

05

Visual Inspection

Examine fleet maps, battery profiles, and route schedules

The dashboard serves as the central control panel for experimental design, execution monitoring, and solution quality assessment.

Effect of Parameter Changes

Population Size and Diversity

Larger populations increase genetic diversity, potentially improving solution quality but requiring more computational time. Trade-offs must balance exploration breadth against runtime efficiency.

Crossover Probability

Higher values facilitate combination of beneficial route fragments, enhancing exploration of solution space but potentially disrupting good schemata.

Mutation Probability

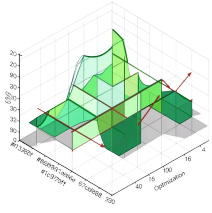
Increased mutation prevents premature convergence to local optima, though excessive mutation destroys accumulated solution quality.

Balance Required

Optimal results emerge from balanced parameter settings rather than maximising individual parameters independently.

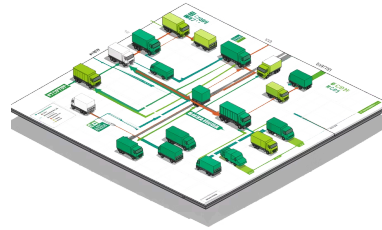
Sensitivity analysis reveals that population size around 100 individuals, crossover probability of 0.8, and mutation probability of 0.1 provides robust performance across benchmark instances.

Dashboard Visualisations



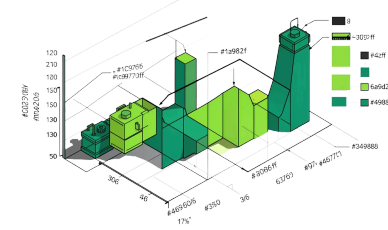
GA Convergence Curve

Best and average fitness values across generations reveal optimisation progress and convergence behaviour



Live Fleet Map

Geometric route visualisation showing depot location, customer positions, and charging station placements



Battery Profile

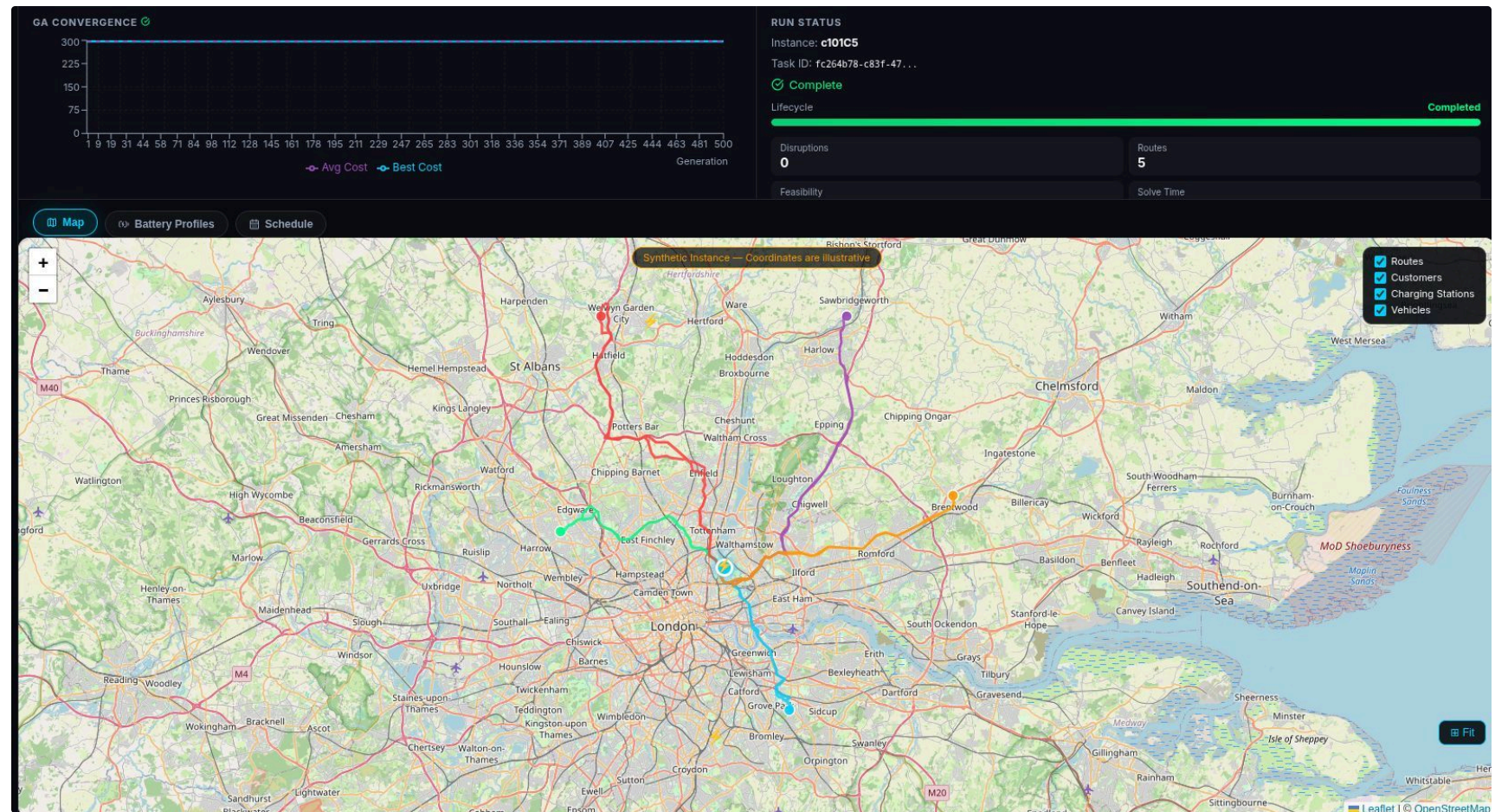
Energy consumption patterns over time, highlighting charging events and discharge segments

Route Schedule Display

Chronological sequence showing vehicle assignments, customer service times, and charging durations with temporal constraints.

Run Status Panel

Real-time indicators showing iteration count, elapsed time, feasibility status, and convergence metrics.



Conclusion

Developed an integrated GA-LP system for energy-constrained dynamic vehicle routing with simple visualizations of the routes, battery profiles and schedules for each vehicle.

Future Phases



Real-Time Scheduling

Dynamic optimisation for fleet operations responding to live demand



Advanced Disruption Handling

Delays, failures, and route changes with recovery procedures



WebSocket Updates

Live telemetry for run status and fleet monitoring



Database Persistence

Schedules, disruptions, and historical run storage



Production Hardening

Deployment for operational environments

The current implementation will extend into a comprehensive dynamic scheduling and disruption-aware decision support system, integrating real-time optimisation with robust recovery procedures for electric vehicle fleet management

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